

JCars Logistics Power BI Final Project Documentation

Data preparation, modelling, analysis, validation and report delivery

Item	Final value
Student	Jack Kisutsa
Programme	Data Analytics, Data Science and AI
Institution	LuxDev HQ Academy
Project	JCars Logistics Power BI Business Intelligence Assessment
Source dataset	Jcars_data.csv
Source records	276 rows and 32 columns
Reporting period	2025 to 2026
Completion date	23 September 2026
Submission deadline	26 September 2026 at 4:00 PM EAT
Document status	Final project record

This document records the complete analytical workflow and the decisions needed to reproduce and review the Power BI solution.

1 Executive summary

The project transformed a deliberately inconsistent vehicle sales and logistics dataset into a validated Power BI analytical model. The solution preserves all 276 source records, standardises commercial and operational fields, converts monetary values to Kenya shillings, separates active sales from returned, cancelled and refunded transactions, and exposes data-quality limitations rather than hiding them.

The final report contains an Executive Overview, Sales and Profitability, Operations and Logistics, Customer and Data Quality, Insights, a Branch Details drill-through page and a Vehicle Tooltip page. All refresh, filter, navigation, drill-through, tooltip and visual-quality checks passed before finalisation.

Headline result	Final value	Interpretation
Total source orders	276	All source records retained for auditability
Active sales orders	153	Orders included in active commercial performance
Active sales revenue	KES 905.0M	Revenue from active transactions
Active sales cost	KES 804.7M	Cost associated with active sales
Active gross profit	KES 99.2M	Profit after the documented cost treatment
Active gross margin	11.65%	Active gross profit relative to analysis revenue
On-time delivery rate	91.87%	113 of 123 delivered orders with valid durations
Return rate	34.38%	88 of 256 orders with known return status
Revenue match rate	48.54%	116 matches among 239 records that could be validated
Date issue orders	48	Missing or invalid dates and delivery-before-order cases

2 Project background and objective

JCars Logistics imports, sells and delivers vehicles through branches in Kenya. Management needs a concise view of revenue, profitability, branch and vehicle performance, logistics efficiency, customer outcomes, returns and data reliability. The assignment required a complete Power BI solution from raw-data investigation through business recommendations and publication-ready documentation.

2.1 Dataset grain

The analysis treats each source row as one order record or order line associated with one customer, vehicle configuration, sales representative, branch and delivery outcome. Units Sold can exceed one, so an order record is not necessarily one vehicle. Source Row ID is the audit key; Order ID remains a business identifier because the source contains blanks, placeholders and duplicates.

3 Requirements completion register

Requirement	Status	Evidence
Data-quality audit with at least 10 issues	Complete	21 significant issue categories documented
Power Query cleaning and preparation	Complete	Column-by-column transformations and flags implemented
Currency identification and KES conversion	Complete	KES, USD, EUR, ZAR and question-mark prefixes treated consistently
Validation of cleaned data	Complete	Row counts, types, dates, revenue, model and interactions checked

Requirement	Status	Evidence
Analytical data model	Complete	FactSalesOrders plus four dimensions and a measure table
Date table	Complete	730 dates covering 2025 and 2026 and marked as a Date table
Purposeful DAX measures	Complete	Commercial, profitability, delivery, returns and quality measures
Executive Dashboard as Page 1	Complete	Executive Overview page
Detailed analysis pages	Complete	Three detailed analytical pages plus an insights page
Slicers and interactions	Complete	Year, region, branch and vehicle type synchronised across main pages
Drill-through	Complete	Branch Details page
Report tooltip	Complete	Vehicle Tooltip page
Page navigation	Complete	Page Navigator across visible pages
Five analyst-defined questions	Complete	Five questions documented and answered
Five management insights	Complete	Five evidence-supported insights
Three management recommendations	Complete	Five recommendations linked to findings
Final report testing	Complete	Refresh, totals, slicers, interactions, drill-through and tooltip passed
GitHub README	Prepared	Final Markdown file produced for repository upload
Technical article	Prepared	Final Markdown draft produced for Dev.to publication
Public links and submission email	Pending publication	Complete after GitHub upload and Dev.to publication

4 Initial data-quality audit

The raw CSV contained 276 records and 32 fields. Profiling showed no completely identical rows, but nearly every analytical area contained format, category, validity or completeness problems. The cleaning approach retained uncertain records with flags or null values rather than silently deleting them.

ID	Area	Observed issue	Treatment
DQ01	Order identifiers	Blanks, placeholders and duplicate business IDs	Use Source Row ID; preserve and flag source IDs
DQ02	Order dates	Mixed formats, Excel serials, blanks and invalid dates	Controlled parsing and date-quality flags
DQ03	Delivery dates	Mixed formats, serials, blanks and impossible dates	Controlled parsing; unsupported values become null
DQ04	Date chronology	Delivery dates occurring before order dates	Flag and exclude from duration averages
DQ05	Customer type	Spelling, case and synonym variants	Map to controlled categories
DQ06	Customer age	Negative, zero, extreme, text and blank values	Parse plausible ages; flag invalid or not applicable
DQ07	Geography	Region, county and city variants	Standardise spelling and abbreviations
DQ08	Branch	Yard, branch and location variants	Map to eight standard branches
DQ09	Sales representative	Typos, digits and spacing inconsistencies	Clean and map to ten standard names
DQ10	Lead source	Typos, case and synonym variants	Map to controlled lead-source values

ID	Area	Observed issue	Treatment
DQ11	Vehicle descriptions	Make and model spelling inconsistencies	Standardise make and model
DQ12	Vehicle attributes	Vehicle type, fuel, transmission and colour variants	Standardise category values
DQ13	Vehicle year	Text, malformed, implausible and future years	Parse valid years and flag exceptions
DQ14	Units sold	Words, labels, blanks, zero and negative values	Convert supported text; null or flag invalid quantities
DQ15	Monetary formats	Currency labels, commas, M suffixes and text tokens	Separate currency and numeric components
DQ16	Mixed currencies	KES, USD, EUR, ZAR and unknown-looking prefixes	Apply documented fixed conversion rates
DQ17	Discount	Percent signs, decimals, whole numbers and invalid rates	Standardise to a decimal rate
DQ18	Recorded revenue	Missing, negative, zero and inconsistent amounts	Calculate independent revenue and classify variance
DQ19	Operational status	Payment and delivery status synonyms	Map to controlled transaction status categories
DQ20	Ratings and reviews	Text ratings and invalid or negative counts	Extract valid numeric values and flag others
DQ21	Returned indicator	Boolean, yes/no, letters, text and blanks	Map to True, False or Unknown

5 Power Query cleaning and transformation

The raw query was preserved as Raw JCars Data with load disabled. Stg JCars Data referenced the raw query and became the reusable cleaning layer. Transformations were completed column by column, with dates revisited after the other fields were standardised. Fact and dimension queries were then created from the prepared data.

Stage	Area	Completed transformation
1	Source control	Imported CSV, promoted headers, retained 276 rows and preserved the raw query
2	Audit key	Added Source Row ID from 1 to provide a stable row-level identifier
3	Text preparation	Applied trimming, cleaning and consistent null handling before mappings
4	Order ID	Standardised prefixes and case while retaining Source Row ID as the reliable audit key
5	Customer	Cleaned names, mapped customer types and validated customer ages
6	Geography	Standardised region, county, city and branch values
7	Sales and leads	Standardised representative names and lead sources
8	Vehicle	Standardised make, model, type, year, fuel type, transmission and colour
9	Quantity	Converted supported Units Sold values to whole numbers and flagged invalid quantities
10	Money parsing	Extracted currency prefixes, separators, decimal amounts and million multipliers
11	KES conversion	Applied fixed rates to selling price, cost, delivery fee, logistics cost and recorded revenue
12	Discount	Converted percentage and decimal representations to a decimal rate
13	Operational fields	Standardised payment method, payment status and delivery status
14	Customer outcome	Parsed rating, review count and returned status
15	Dates	Parsed Excel serials and recognised text formats, corrected documented impossible dates and retained unsupported values as null

Stage	Area	Completed transformation
16	Calculations	Created revenue, cost, gross profit, delivery days and validation fields
17	Classification	Created Final Transaction Status, Date Validation Status and Revenue Validation Status
18	Model outputs	Created FactSalesOrders, DimBranch, DimSalesRep, DimVehicle and DimDate

5.1 Core row-level calculations

Calculated Revenue KES was created in Power Query for row-level validation:

$[Units\ Sold] * [Unit\ Selling\ Price\ KES] * (1 - [Discount\ Rate]) + [Delivery\ Fee\ KES]$

Delivery Days was calculated only when both dates were valid and the delivery date was not earlier than the order date:

$Duration.Days([Delivery\ Date] - [Order\ Date])$

Revenue Validation Status classified each record as Match, Mismatch or Cannot Validate. Date Validation Status classified missing or invalid order dates, missing or invalid delivery dates, delivery-before-order records and valid sequences.

6 Date cleaning decisions and exceptions

Dates used controlled parsing because the source mixed Excel serials, ISO-like text, month names, slashes and impossible calendar dates. Corrections were made only where surrounding evidence supported a reasonable interpretation. Unsupported values remained null.

Source row	Source condition	Final treatment	Reason
26	Order 16/04/2025; Delivery 05/04/2025	Retain parsed values and flag Delivery Before Order Date	Both dates are valid but chronology is inconsistent
50	Order 09-Sept-25	Parse as 09/09/2025	Recognised month-name format
79	Delivery April 31 2026	Correct to 30/04/2026	April has 30 days and a valid order date exists
155	Delivery 31/02/2026; Order missing	Leave Delivery Date null	No supporting order date for a reliable correction
173	Order 31/02/2026	Correct to 28/02/2026	February 2026 ends on the 28th and delivery evidence exists
189	Order 01/02/2025; Delivery 05-Jan-25	Interpret Order Date as 02/01/2025	Day/month interpretation avoids an unsupported delivery-before-order sequence
195	Delivery 31/02/2026	Correct to 28/02/2026	Valid order date supports end-of-February interpretation
201	Order April 31 2026; Delivery unavailable	Leave Order Date null	No delivery date or other evidence to support correction
209	Delivery April 31 2026	Correct to 30/04/2026	April has 30 days and a valid order date exists
210	Order 24-Sept-25	Parse as 24/09/2025	Recognised month-name format
230	Order 03/06/2026; Delivery 31/05/2026	Retain and flag Delivery Before Order Date	Both source dates are valid
237	Excel serial dates produce delivery before order	Retain and flag Delivery Before Order Date	Serials are valid but chronology is inconsistent
258	Order 01/12/2026; Delivery serial 46042	Interpret Order Date as 12/01/2026	Chronology and ambiguous slash format

Source row	Source condition	Final treatment	Reason
			support month/day interpretation
261	Order 2026-13-04; delivery earlier after parsing	Flag for date-quality review	Malformed source format and chronology require caution
270	Order April 31 2026; valid Delivery 22/04/2026	Leave Order Date null and document	Changing to 30 April would still place delivery before order

7 Currency standardisation

All monetary values were converted to KES before aggregation. Values without a currency label were treated as KES, as required by the assessment. The model uses fixed rounded project rates to make every transformation reproducible across the 2025-2026 dataset. These rates are analytical assumptions, not transaction-date accounting rates.

Source currency	Project rate to KES	Treatment
KES or KSh	1.00	Retained as Kenya shillings
No currency label	1.00	Assumed KES as required by the assessment
USD or \$	130.00	Converted to KES
EUR	150.00	Converted to KES
ZAR or R	7.50	Converted to KES
Question-mark prefix	130.00	Treated as USD as a documented dataset-specific assumption

External reference: Central Bank of Kenya Foreign Exchange Rates, accessed 23 September 2026. The CBK publishes indicative daily rates based on average buying and selling rates of market participants. On the access date, the displayed indicative rates included USD/KES 129.45 and EUR/KES 147.92, supporting the reasonableness of the rounded fixed rates used for this educational model. Source: <https://www.centralbank.go.ke/rates/forex-exchange-rates/>

Limitation: using one fixed rate removes daily exchange-rate effects. The dashboard therefore compares operational performance on a common analytical basis but should not be treated as a replacement for transaction-date financial accounting.

8 Assumptions and business rules

Topic	Final rule
Dataset grain	One row is one order record or order line; Source Row ID is the audit key.
Unlabelled money	Treat as KES in accordance with the assessment instructions.
Question-mark currency	Treat as USD because the affected amounts follow the foreign-currency scale; document as an assumption.
Calculated revenue	Units Sold x Unit Selling Price KES x (1 - Discount Rate) + Delivery Fee KES.
Active sale	Final Transaction Status equals Active Sale. Returned, cancelled and refunded transactions are excluded from active commercial KPIs.
Gross profit	Use the prepared row-level gross-profit field and aggregate it through explicit measures.
Loss-making order	An active order with calculated gross profit below zero.

Topic	Final rule
Delivery duration	Use only records with valid Order Date and Delivery Date and Delivery Date greater than or equal to Order Date.
On-time delivery	Valid delivered order completed within 30 days.
Returned order	Returned flag equals True; blank or unknown values are excluded from the return-rate denominator.
Revenue match	Calculated and recorded revenue meet the project validation tolerance and both values are available.
Unknown values	Retain as null or Unknown and expose through quality measures rather than deleting the record.
Date corrections	Correct impossible end-of-month values only where another valid date supports the decision; otherwise leave null.

9 Data model

The final model uses a compact star schema. FactSalesOrders contains the order-level commercial, logistics, status and data-quality fields. Dimensions provide consistent filtering, while the _Measures table stores explicit reusable DAX measures.

Table	Rows	Purpose	Key
FactSalesOrders	276	Order-level fact table	Source Row ID and dimension foreign keys
DimDate	730	Continuous calendar for 2025-2026	Date
DimBranch	8	Branch and geography attributes	Branch Key
DimSalesRep	10	Standard sales-representative names	Sales Rep Key
DimVehicle	35	Vehicle make, model and type combinations	Vehicle Key
_Measures	N/A	Central store for explicit DAX measures	Not applicable

Fact field	Dimension field	Status	Reason
FactSalesOrders[Order Date]	DimDate[Date]	Active, many-to-one	Primary sales-date analysis
FactSalesOrders[Delivery Date]	DimDate[Date]	Inactive, many-to-one	Activated in delivery-date measures
FactSalesOrders[Branch Key]	DimBranch[Branch Key]	Active, many-to-one	Branch and geography filtering
FactSalesOrders[Sales Rep Key]	DimSalesRep[Sales Rep Key]	Active, many-to-one	Representative filtering
FactSalesOrders[Vehicle Key]	DimVehicle[Vehicle Key]	Active, many-to-one	Vehicle filtering

10 Important DAX measures

The following measures form the core reporting layer. Names match the report wherever possible.

Measure	Representative DAX	Purpose
Total Orders	COUNTROWS(FactSalesOrders)	All source order records

Measure	Representative DAX	Purpose
Active Orders	CALCULATE([Total Orders], FactSalesOrders[Final Transaction Status] = "Active Sale")	Active transaction count
Active Sales Revenue KES	CALCULATE(SUM(FactSalesOrders[Calculated Revenue KES]), FactSalesOrders[Final Transaction Status] = "Active Sale")	Active revenue
Active Sales Cost KES	CALCULATE(SUM(FactSalesOrders[Calculated Cost KES]), FactSalesOrders[Final Transaction Status] = "Active Sale")	Active cost
Active Gross Profit KES	CALCULATE(SUM(FactSalesOrders[Calculated Gross Profit KES]), FactSalesOrders[Final Transaction Status] = "Active Sale")	Active profitability
Active Gross Margin	DIVIDE([Active Gross Profit KES], [Profit Analysis Revenue KES])	Profitability ratio
Loss-Making Orders	COUNTROWS(FILTER(FactSalesOrders, FactSalesOrders[Final Transaction Status] = "Active Sale" && FactSalesOrders[Calculated Gross Profit KES] < 0))	Active loss-making orders
On-Time Delivery Rate	DIVIDE([On-Time Delivered Orders], [Delivered Orders with Valid Duration])	Delivery service level
Return Rate	DIVIDE([Returned Orders], [Orders with Known Return Status])	Known-status return risk
Revenue Match Rate	DIVIDE([Revenue Match Orders], [Revenue Match Orders] + [Revenue Mismatch Orders])	Validation success rate
Orders by Delivery Date	CALCULATE([Total Orders], USERELATIONSHIP(FactSalesOrders[Delivery Date], DimDate[Date]))	Delivery-date analysis
Active Revenue YTD	TOTALYTD([Active Sales Revenue KES], DimDate[Date])	Cumulative active revenue

11 Validation results

Validation check	Final evidence	Result
Source row count	276 rows retained	Passed
Data types	Date, whole number, decimal, percentage and text fields reviewed	Passed
Date table	730 unique dates, marked as the Date table	Passed
Date quality	48 problematic records exposed through Date Validation Status	Passed with disclosed issues
Revenue validation	116 matches, 123 mismatches and 37 cannot validate	Passed with disclosed issues
Transaction status	153 active, 73 returned, 42 cancelled and 8 refunded	Passed
Model integrity	Dimension relationships configured with single-direction filtering	Passed
Headline totals	All baseline KPI values reconciled after slicers were cleared	Passed
Refresh	Model refreshed without Power Query errors	Passed
Interactivity	Slicers, chart interactions, page navigation, drill-through and tooltip tested	Passed
Visual quality	Titles, labels, KES formats, percentages, blanks and chronology reviewed	Passed

12 Report structure and analytical flow

Page	Purpose	Main content
Executive Overview	High-level business condition	Revenue, profit, margin, active orders, on-time delivery, rating and core comparisons
Sales and Profitability	Commercial performance	Revenue, cost, profit, margin, loss-making orders, sales representatives, discounts and vehicles

Page	Purpose	Main content
Operations and Logistics	Delivery and fulfilment	Delivered orders, duration, on-time rate, delivery status and logistics cost
Customer and Data Quality	Customer outcomes and reliability	Ratings, returns, revenue validation and data-quality issues
Insights	Management interpretation	Business questions, findings, insights and recommendations
Branch Details	Drill-through investigation	Selected branch KPIs, trend, sales representatives and order-level details
Vehicle Tooltip	Context without overcrowding	Vehicle revenue, profit, margin and return rate

Year, Region, Branch and Vehicle Type slicers were synchronised across the four analytical pages. A Page Navigator connects the visible pages. Branch Details and Vehicle Tooltip are hidden support pages. Cross-filtering allows users to move from a high-level pattern to its contributing categories.

13 Final report evidence

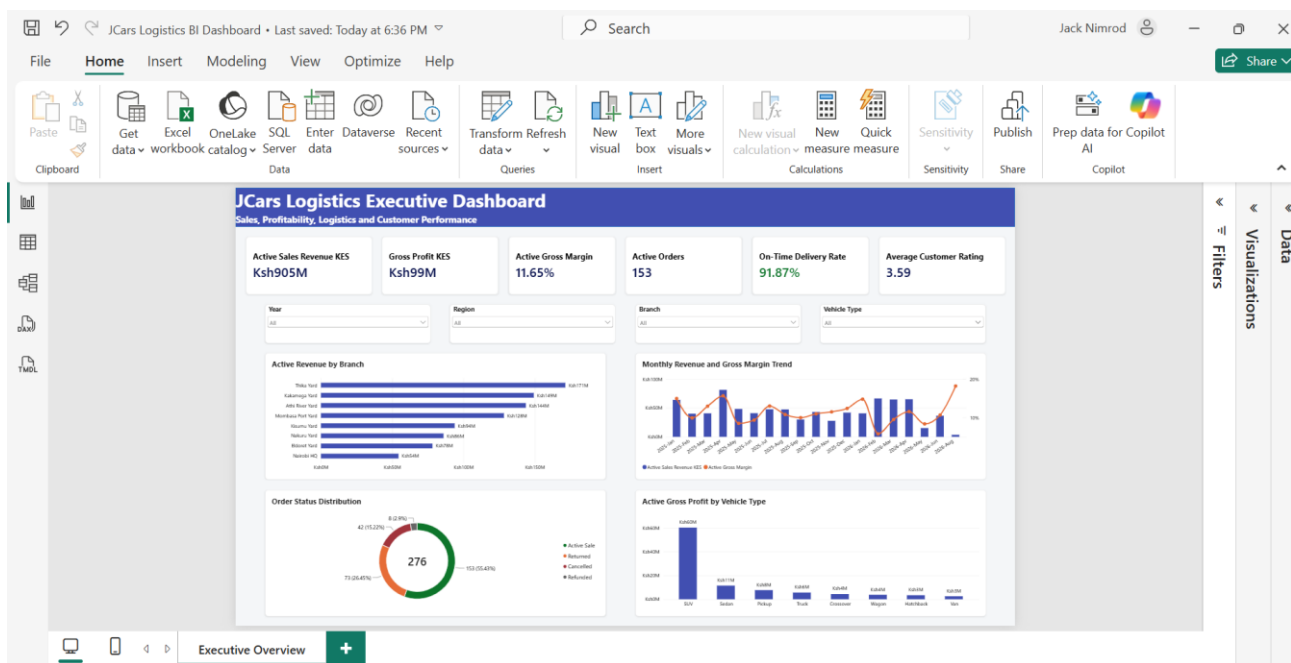


Figure 1 Executive Overview

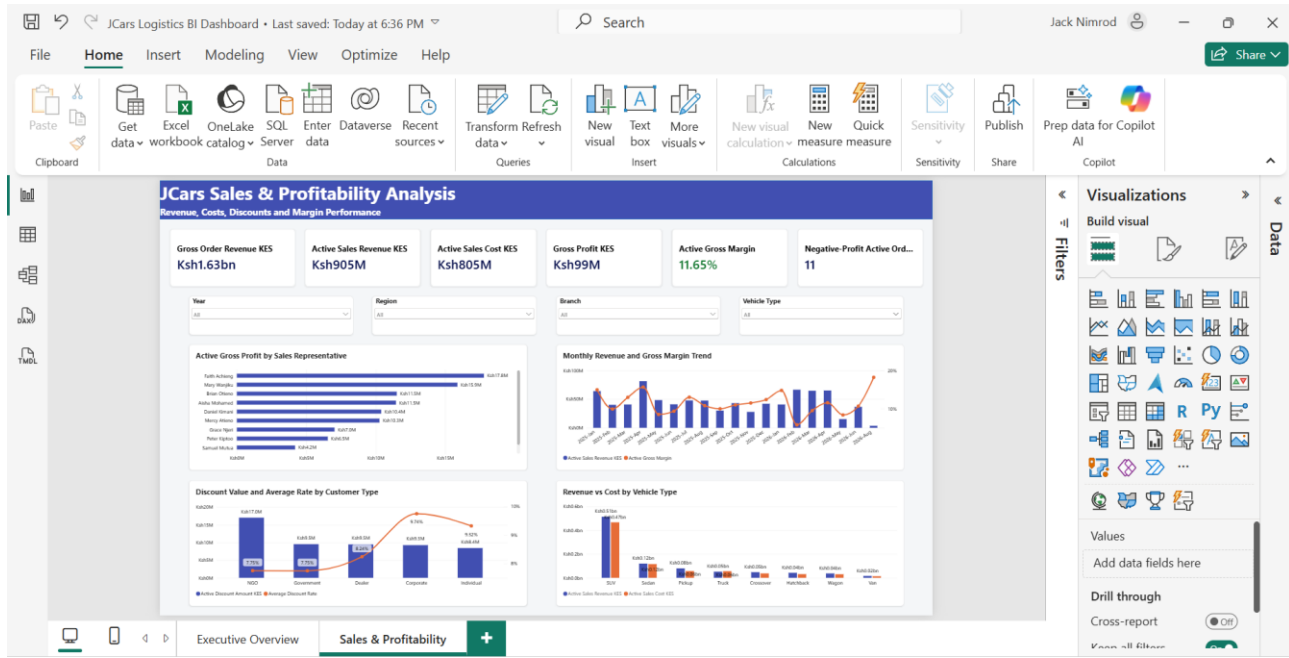


Figure 2 Sales and Profitability

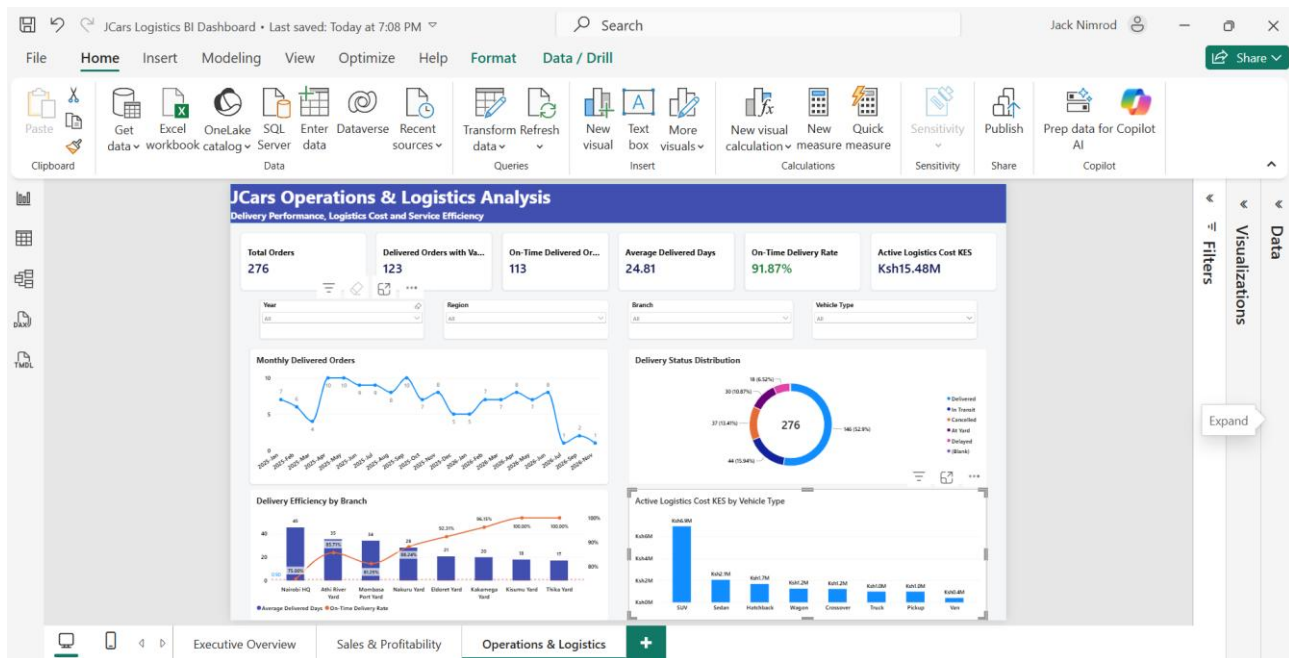


Figure 3 Operations and Logistics

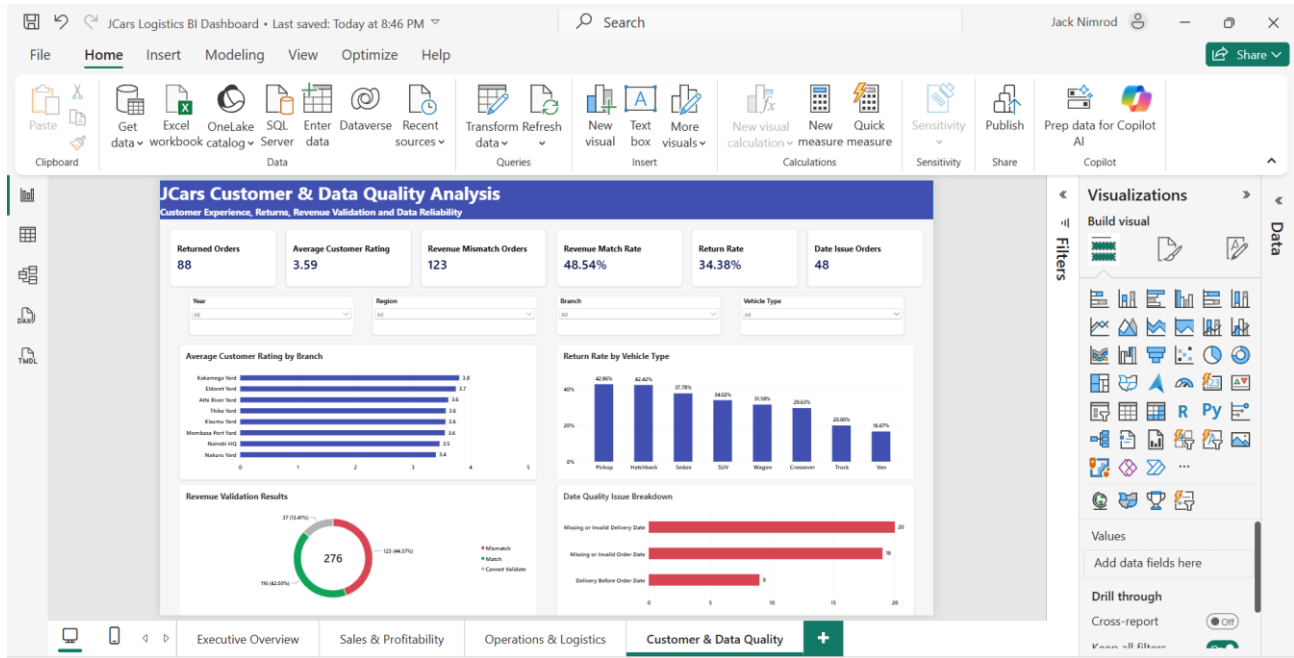


Figure 4 Customer and Data Quality

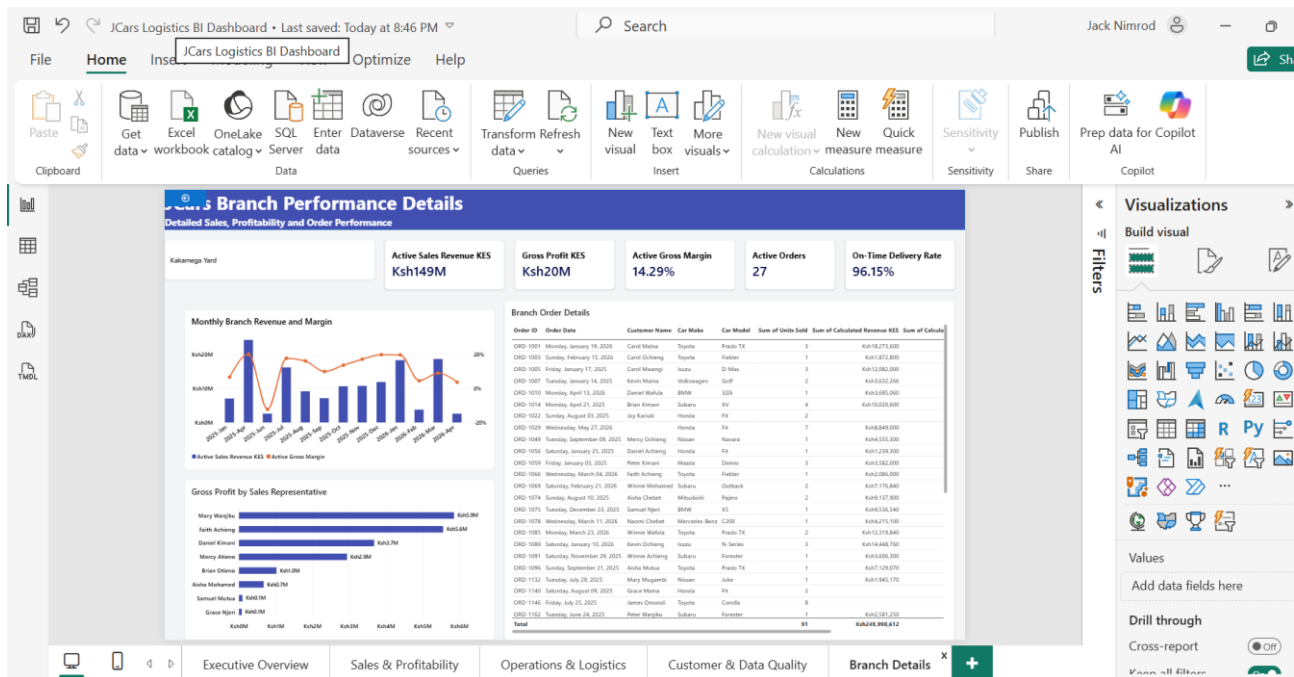


Figure 5 Branch Details drill-through

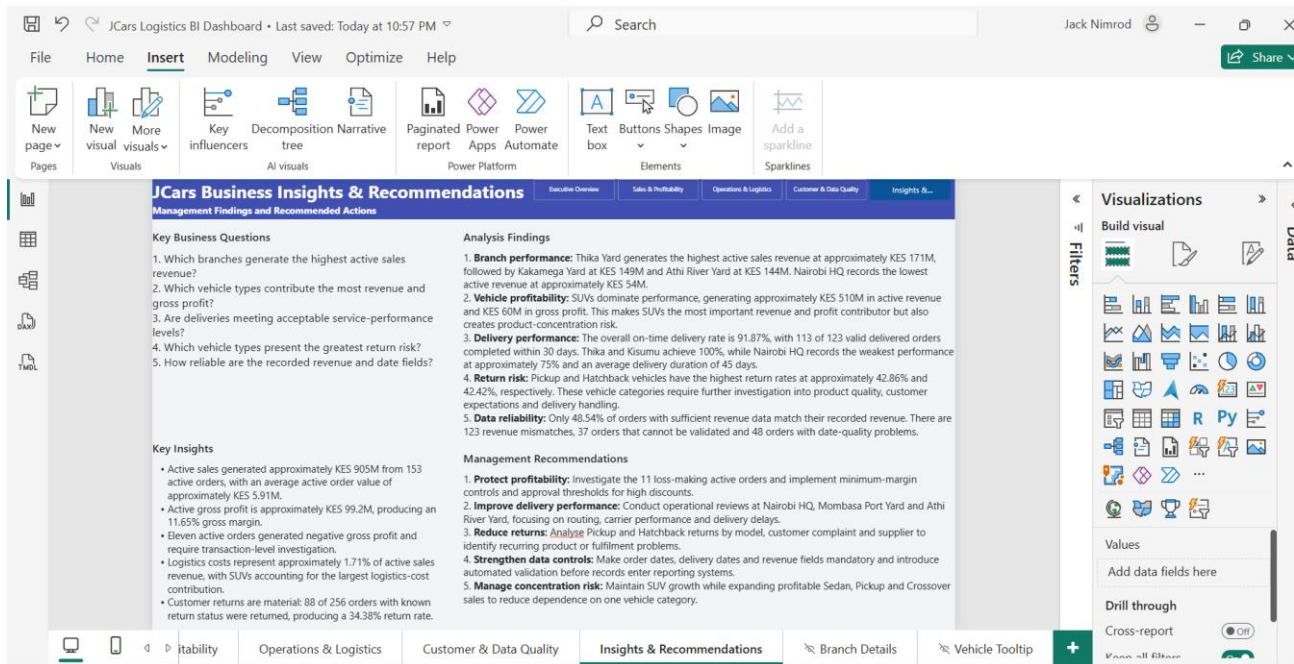


Figure 6 Business Insights and Recommendations

14 Analyst-defined business questions

Question	Why it matters
Which branches generate the highest active sales revenue?	Branch revenue ranking identifies commercial concentration and underperformance.
Which vehicle types contribute the most revenue and gross profit?	Vehicle contribution supports stock, procurement and portfolio decisions.
Are deliveries meeting acceptable service-performance levels?	Delivery performance affects customer experience and operational cost.
Which vehicle types present the greatest return risk?	Return concentration can reveal product, fulfilment or expectation problems.
How reliable are the recorded revenue and date fields?	Management should understand whether source limitations affect confidence in reported results.

15 Findings and management insights

Finding	Evidence-supported interpretation
Branch concentration	Thika Yard led active revenue at approximately KES 171M, followed by Kakamega Yard at KES 149M and Athi River Yard at KES 144M. Nairobi HQ recorded approximately KES 54M.
Vehicle concentration	SUVs generated approximately KES 510M in active revenue and KES 60M in gross profit, making them the dominant contributor and the largest concentration exposure.
Delivery performance	The overall on-time rate was 91.87%. Thika and Kisumu reached 100%, while Nairobi HQ recorded about 75% and an average delivery duration near 45 days.
Return risk	Pickup and Hatchback recorded the highest return rates at approximately 42.86% and 42.42%.
Data reliability	Only 48.54% of validateable orders matched recorded revenue. There were 123 mismatches, 37 records that could not be validated and 48 date-quality issues.

Finding	Evidence-supported interpretation
Commercial scale	Active sales produced approximately KES 905M from 153 orders, with an average active order value near KES 5.91M.
Profitability	Active gross profit was approximately KES 99.2M at an 11.65% margin; 11 active orders were loss-making.
Logistics exposure	Active logistics cost was approximately KES 15.48M, or 1.71% of active revenue. SUVs accounted for the largest absolute logistics cost.
Returns	Eighty-eight of 256 orders with known return status were returned, producing a 34.38% return rate.

16 Management recommendations

Priority	Recommended action	Monitoring evidence
Protect profitability	Investigate the 11 loss-making active orders and implement minimum-margin controls and approval thresholds for high discounts.	Loss-Making Orders and Active Gross Margin
Improve delivery performance	Review routing, carrier performance and delay causes at Nairobi HQ, Mombasa Port Yard and Athi River Yard.	Branch delivery efficiency and Average Delivered Days
Reduce returns	Analyse Pickup and Hatchback returns by model, complaint, supplier and delivery handling before changing inventory decisions.	Return Rate by Vehicle Type
Strengthen data controls	Make order date, delivery date, currency and revenue fields mandatory and validate them before records enter reporting systems.	Revenue Match Rate and Date Issue Orders
Manage concentration risk	Maintain profitable SUV growth while expanding profitable Sedan, Pickup and Crossover opportunities to reduce dependence on one vehicle category.	Revenue and profit by vehicle type

17 Challenges and lessons learned

Challenge	Resolution	Lesson
Mixed dates and invalid calendar values	Controlled parsing, documented corrections and explicit quality flags were safer than bulk type conversion.	Ambiguous dates need evidence-based rules and an audit trail.
Mixed currencies and formats	Currency and numeric parsing were separated before applying fixed rates.	Standardise units before aggregation.
High mismatch volume	Calculated revenue was retained alongside recorded revenue with a validation status.	A reconciliation result is itself an analytical finding.
Reserved measure-table name	The table was named _Measures rather than Measures.	Power BI object naming rules should be considered early.
Date functions and M/DAX differences	Unsupported Date.Compare logic was replaced with direct comparisons and null checks.	Power Query M and DAX have different functions and contexts.
Card and navigator labels	Measure and page names were shortened where automatic labels were inherited.	Some visuals inherit semantic-model names and cannot be relabelled independently.
Percentage target line	The optional 90% constant line was omitted after it attached to the wrong axis.	A visual feature should be retained only when it communicates the intended metric correctly.

18 Final quality assurance

- Saved the final PBIX under a clear final filename and retained a working backup.

- Refreshed the model successfully and confirmed that all pages rendered.
- Cleared all slicers and reconciled baseline KPI values.
- Tested Year, Region, Branch and Vehicle Type slicers on the main analysis pages.
- Tested all navigation buttons and confirmed that labels were readable.
- Tested Branch Details drill-through and the back-navigation experience.
- Tested the custom Vehicle Tooltip on multiple vehicle categories.
- Tested chart cross-filtering and cleared all selections afterward.
- Reviewed the Filters pane for unintended visual, page or report filters.
- Checked titles, KES labels, percentages, chronological month sorting, blank categories and layout consistency.
- Confirmed that Branch Details and Vehicle Tooltip remained hidden support pages.
- Exported and reviewed the final report PDF.

19 GitHub and publication plan

Recommended repository structure:

```
JCars-Logistics-Analysis/
├── README.md
├── JCars_Logistics_Power_BI_Dashboard_Final.pbix
├── data/Jcars_data.csv
├── documentation/JCars_Logistics_Power_BI_Final_Project_Documentation.pdf
├── documentation/JCars_Logistics_Data_Cleaning_Log.xlsx
├── screenshots/
│   ├── executive-overview.png
│   ├── sales-profitability.png
│   ├── operations-logistics.png
│   ├── customer-data-quality.png
│   ├── branch-details.png
│   └── insights-recommendations.png
```

The repository README and technical article drafts accompany this document. After upload, verify the repository in a private browser window, publish the article on Dev.to, test both links and submit them by email with the student name, cohort and track.

20 Final submission checklist

Submission item	Status
Power BI report and all quality checks	Complete
Final project documentation	Complete
Excel cleaning and validation log	Complete
GitHub README draft	Complete
Dev.to article draft	Complete
PBIX and supporting files uploaded to GitHub	Pending user upload
Dev.to article published	Pending user publication
GitHub and Dev.to links tested publicly	Pending publication
Submission email sent before deadline	Pending final submission